

Enchantment::doPostHurt and Enchantment::doPostAttack are called twice for players

MC-248272 Enchantments

Status	Resolved / Fixed
Affected	1.18.1, 22w03a, 22w05a, 1.18.2, 1.19.2, 1.19.4 Pre-release 3 (+2 more)
Fixed in	24w18a
Reported	2022-01-26

STEPS TO REPRODUCE

1. Launch Minecraft Java Edition 1.20.1.
2. Create a new Creative world with default settings.
3. Obtain a sword and apply Bane of Arthropods to it using `/enchant @p minecraft:bane_of_arthropods 1``.
4. Attack a target for which Bane of Arthropods applies.
5. Observe that the target receives the Slowness effect twice.
6. Obtain armor and apply Thorns to it using `/enchant @p minecraft:thorns 1``.
7. Have another player attack the player wearing the Thorns-enchanted armor.
8. Observe that Thorns damages the attacker twice and reduces the armor's durability twice.

OBSERVED BEHAVIOUR

In Minecraft Java Edition 1.18.1 through 1.20.1, `EnchantmentHelper.doPostHurtEffects` and `EnchantmentHelper.doPostDamageEffects` can process the same enchantment twice when a player is involved. Each method first iterates over the entity's complete inventory through `getAllSlots()`, which includes the main-hand item, and then performs another iteration over the main-hand item when the relevant entity is a player. As a result, `Enchantment.doPostHurt` or `Enchantment.doPostAttack` may be invoked twice for the same enchanted item during a single interaction. This causes enchantment effects to be applied twice, such as Bane of Arthropods applying Slowness twice and Thorns damaging the attacker twice while also reducing the durability of the Thorns item twice.

EXPECTED BEHAVIOUR

Enchantment post-hurt and post-attack callbacks should be invoked only once for a player during each relevant combat event. Bane of Arthropods should apply Slowness once, while Thorns should damage the attacker once and reduce the enchanted item's durability only once. Other enchantment effects should likewise not be duplicated because the player's inventory and equipped item are processed redundantly.

ENVIRONMENT

Minecraft Java Edition Versions: 1.18.1, 22w03a, 22w05a, 1.18.2, 1.19.2, 1.19.4 Pre-release 3, 1.20.1, 1.20.2 (includes snapshots)

ORIGINAL DESCRIPTION

The `Enchantment::doPostHurt` and `Enchantment::doPostAttack` functions are called twice for the player causing some bugs.

- The Bane of Arthropods enchantment gives the Slowness effect twice
- The Thorns enchantment deals damage to the attacker and the durability of the item with this enchantment twice.

The problem is that the `EnchantmentHelper::doPostHurtEffects` and `EnchantmentHelper::doPostDamageEffects` actually call these functions twice:

Vanilla code 1.18.1-1.20.1

```
public static void doPostHurtEffects(LivingEntity defender, Entity attacker) {
    EnchantmentVisitor consumer = (enchantment, lvl) -> enchantment.doPostHurt(defender, attacker, lvl);
    if (defender != null) {
        EnchantmentHelper.runIterationOnInventory(consumer, defender.getAllSlots());
    }
}
```

```

}
if (attacker instanceof Player) {
    EnchantmentHelper.runIterationOnItem(consumer, defender.getMainHandItem());
}
}

public static void doPostDamageEffects(LivingEntity attacker, Entity target) {
    EnchantmentVisitor consumer = (enchantment, lvl) -> enchantment.doPostAttack(attacker, target, lvl);
    if (attacker != null) {
        EnchantmentHelper.runIterationOnInventory(consumer, attacker.getAllSlots());
    }
    if (attacker instanceof Player) {
        EnchantmentHelper.runIterationOnItem(consumer, attacker.getMainHandItem());
    }
}

Fix suggestion

public static void doPostHurtEffects(LivingEntity defender, Entity attacker) {
    EnchantmentVisitor consumer = (enchantment, lvl) -> enchantment.doPostHurt(defender, attacker, lvl);
    if (defender != null) {
        EnchantmentHelper.runIterationOnInventory(consumer, defender.getAllSlots());
    } else if (attacker instanceof Player) {
        EnchantmentHelper.runIterationOnItem(consumer, defender.getMainHandItem());
    }
}

public static void doPostDamageEffects(LivingEntity attacker, Entity target) {
    EnchantmentVisitor consumer = (enchantment, lvl) -> enchantment.doPostAttack(attacker, target, lvl);
    if (attacker != null) {
        EnchantmentHelper.runIterationOnInventory(consumer, attacker.getAllSlots());
    } else if (attacker instanceof Player) {
        EnchantmentHelper.runIterationOnItem(consumer, attacker.getMainHandItem());
    }
}

```

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-248272>